

■ PortfolioGenie

AI-Powered Developer Portfolio Builder

A platform that helps junior developers create professional, high-performance portfolios by analyzing their GitHub activity and generating optimized content.

React

React Hooks

React Router

Prompt Eng.

GitHub OAuth

SEO Optimization

UX/UI Design

WCAG 2.1 AA

Clean Code

Type	Project Overview Document
Version	1.0 — Initial Release
Date	April 2026

Project Overview

PortfolioGenie is a web platform built specifically for junior and mid-level developers who struggle to present their work professionally online. By connecting to a developer's GitHub account, the platform automatically analyses repositories, commit history, and language usage — then uses AI prompt engineering to generate polished, SEO-optimised portfolio content, eliminating the blank-page problem that holds talented engineers back.

What It Does

- Connects to GitHub via OAuth to pull repos, languages, and commit activity.
- Guides users through a multi-step form to capture projects, skills, and experience.
- Uses AI to write a personalised 'About Me' and optimised project descriptions.
- Renders a live, real-time preview of the portfolio using React Router.
- Exports a clean, maintainable static portfolio ready for deployment.

The Four Core Pillars

■ UX / UI Design

- Follow the Design Thinking process: Empathise → Define → Ideate → Prototype → Test.
- Create user personas representing junior developer archetypes and their pain points.
- Produce wireframes and high-fidelity prototypes prioritising developer aesthetics.
- Maintain WCAG 2.1 AA accessibility across all components and colour contrasts.

■■ Core Logic — React & Hooks

- `useState` for all local UI state: form values, step index, and validation errors.
- `useEffect` for side effects: GitHub API calls, debouncing, and live preview syncing.
- Custom hooks — `useGitHub()`, `useFormStep()`, `usePortfolioState()` — for clean logic separation.
- React Router v6 for seamless programmatic navigation between builder and live preview.

■ Prompt Engineering & AI Integration

- Analyse GitHub repos and commit patterns to infer a developer's coding style.
- Use structured, layered prompts to generate personalised 'About Me' sections.
- Optimise project descriptions for SEO: readability, keyword density, and meta tags.
- Give users full control to regenerate, edit, or lock any AI-generated content.

■ Best Practices & Code Quality

- Enforce ESLint + Prettier with consistent conventions across the codebase.
- Apply atomic design and SOLID principles for maintainable component architecture.
- Write unit and integration tests (Jest + React Testing Library) for all hooks.
- Ensure generated portfolio code is clean, well-commented, and easy to extend.